

Bayesian Learning

Lecture 3 - Multi-parameter models

Mattias Villani

Department of Statistics
Stockholm University



Lecture overview

- **Multiparameter** models
- **Marginalization**
- **Normal model with unknown variance**
- **Multinomial data**
- **Dirichlet distribution**

Marginalization

- Models with **multiple parameters** $\theta_1, \theta_2, \dots$
- Examples: $x_i \stackrel{iid}{\sim} N(\theta, \sigma^2)$; multiple regression ...
- **Joint posterior distribution**

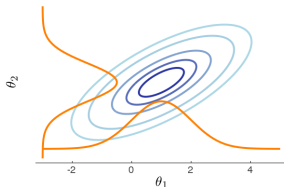
$$p(\theta_1, \theta_2, \dots, \theta_p | y) \propto p(y | \theta_1, \theta_2, \dots, \theta_p) p(\theta_1, \theta_2, \dots, \theta_p).$$

- In vector form

$$p(\boldsymbol{\theta} | \mathbf{y}) \propto p(\mathbf{y} | \boldsymbol{\theta}) p(\boldsymbol{\theta}).$$

- **Marginalize** out parameters. **Marginal posterior** of θ_1 :

$$p(\theta_1 | y) = \int p(\theta_1, \theta_2 | y) d\theta_2 = \int p(\theta_1 | \theta_2, y) p(\theta_2 | y) d\theta_2.$$



Normal model with unknown variance

■ Model

$$x_1, \dots, x_n \stackrel{iid}{\sim} N(\theta, \sigma^2)$$

■ Prior

$$p(\theta, \sigma^2) \propto (\sigma^2)^{-1}$$

■ Posterior

$$\begin{aligned}\theta | \sigma^2, \mathbf{x} &\sim N\left(\bar{x}, \frac{\sigma^2}{n}\right) \\ \sigma^2 | \mathbf{x} &\sim \text{Inv} - \chi^2(n-1, s^2),\end{aligned}$$

where

$$s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}$$

is the usual sample variance.

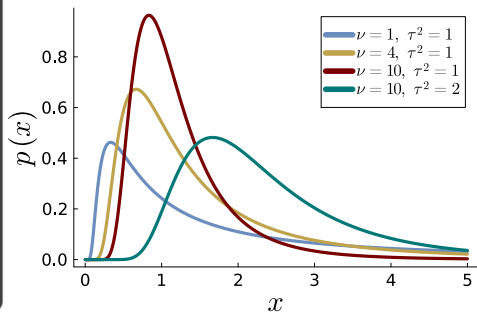
Inv- χ^2 distribution

$X \sim \text{Inv-}\chi^2(\nu, \tau^2), X \in (0, \infty)$

$$p(x) = \frac{(\tau^2 \nu / 2)^{\nu/2}}{\Gamma(\nu/2)} \frac{\exp\left(\frac{-\nu \tau^2}{2x}\right)}{x^{1+\nu/2}}$$

$$\mathbb{E}(X) = \frac{\nu}{\nu - 2} \tau^2$$

$$\mathbb{V}(X) = \frac{2\nu^2 \tau^4}{(\nu - 2)^2(\nu - 4)}$$



Normal model - normal prior

■ Model

$$y_1, \dots, y_n | \theta, \sigma^2 \stackrel{iid}{\sim} N(\theta, \sigma^2)$$

■ Conjugate prior

$$\begin{aligned}\theta | \sigma^2 &\sim N\left(\mu_0, \frac{\sigma^2}{\kappa_0}\right) \\ \sigma^2 &\sim \text{Inv-}\chi^2(\nu_0, \sigma_0^2)\end{aligned}$$

Normal model with normal prior

■ Posterior

$$\begin{aligned}\theta|\mathbf{y}, \sigma^2 &\sim N\left(\mu_n, \frac{\sigma^2}{\kappa_n}\right) \\ \sigma^2|\mathbf{y} &\sim \text{Inv-}\chi^2(\nu_n, \sigma_n^2).\end{aligned}$$

where

$$\begin{aligned}\mu_n &= \frac{\kappa_0}{\kappa_0 + n}\mu_0 + \frac{n}{\kappa_0 + n}\bar{y} \\ \kappa_n &= \kappa_0 + n \\ \nu_n &= \nu_0 + n \\ \nu_n\sigma_n^2 &= \nu_0\sigma_0^2 + (n-1)s^2 + \frac{\kappa_0 n}{\kappa_0 + n}(\bar{y} - \mu_0)^2.\end{aligned}$$

Normal model with normal prior

■ Posterior

$$\begin{aligned}\theta|\mathbf{y}, \sigma^2 &\sim N\left(\mu_n, \frac{\sigma^2}{\kappa_n}\right) \\ \sigma^2|\mathbf{y} &\sim \text{Inv-}\chi^2(\nu_n, \sigma_n^2).\end{aligned}$$

where

$$\begin{aligned}\mu_n &= \frac{\kappa_0}{\kappa_0 + n}\mu_0 + \frac{n}{\kappa_0 + n}\bar{y} \\ \kappa_n &= \kappa_0 + n \\ \nu_n &= \nu_0 + n \\ \nu_n\sigma_n^2 &= \nu_0\sigma_0^2 + (n-1)s^2 + \frac{\kappa_0 n}{\kappa_0 + n}(\bar{y} - \mu_0)^2.\end{aligned}$$

■ Marginal posterior

$$\theta|\mathbf{y} \sim t(\mu_n, \sigma_n^2/\kappa_n, \nu_n)$$

Simulating from posterior - pseudo code

Posterior simulation - iid Gaussian with conjugate prior.

Input: data $\mathbf{x} = (x_1, \dots, x_n)$

number of posterior draws m .

compute μ_n , σ_n^2 , κ_n and ν_n using Figure 50.

for i in $1:m$ **do**

$\sigma^2 \leftarrow \text{RINVCHI2}(\nu_n, \sigma_n^2)$

$\theta \leftarrow \text{RNORMAL}(\mu_n, \sigma^2 / \kappa_n)$

end

Output: m draws for θ and σ^2 from joint posterior.

Function $\text{RINVCHI2}(\nu, \tau^2)$

$x = \text{RCHI2}(\nu)$

$y = \nu \tau^2 / x$

return y

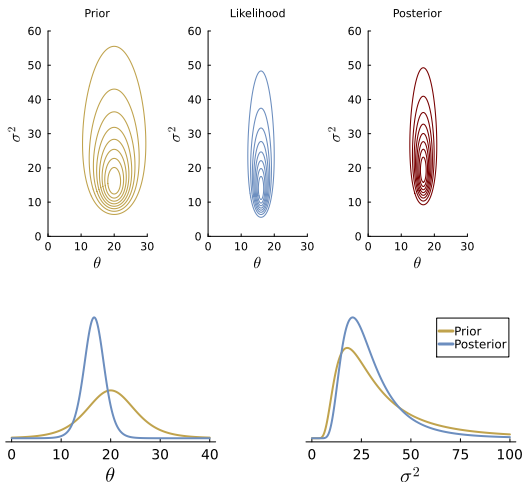
Simulating from posterior - output

draw	θ	σ^2	σ/θ	$\theta \geq 20$
1	18.165	18.451	0.236	0
2	20.431	29.943	0.267	1
3	15.565	29.094	0.346	0
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
10,000	16.400	21.668	0.283	0
Mean	16.645	30.813	0.330	0.066

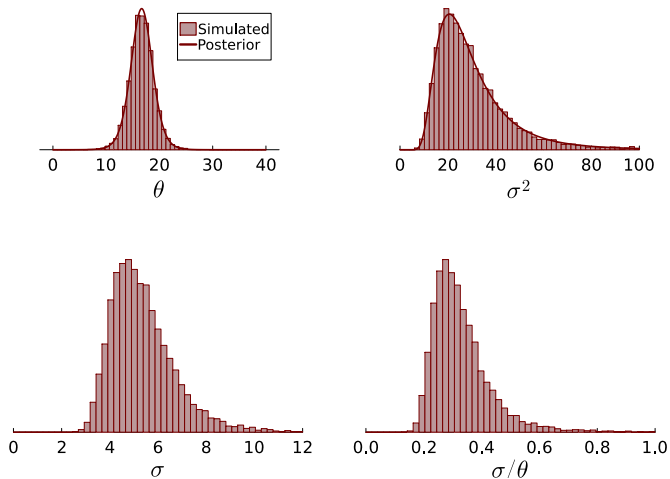
Internet speed data - joint and marginal posteriors

■ Prior:

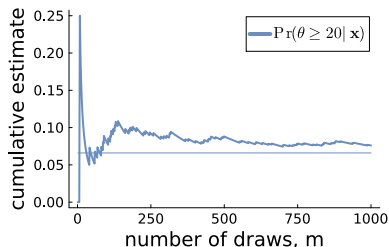
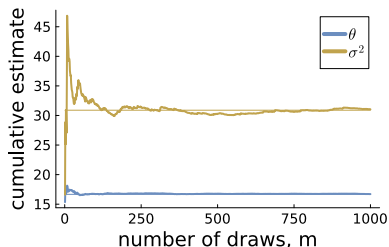
$$\theta | \sigma^2 \sim N\left(20, \frac{\sigma^2}{1}\right) \text{ and } \sigma^2 \sim \text{Inv-}\chi^2(\nu_0 = 5, \sigma_0^2 = 5^2)$$



Monte Carlo simulation



Monte Carlo simulation



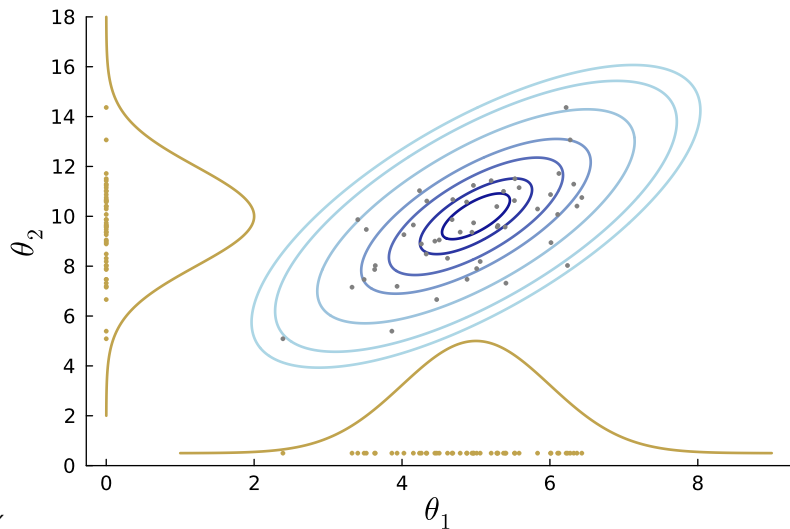
- Law of large numbers for **consistency**:

$$\bar{\theta}_{1:m} \equiv \frac{1}{m} \sum_{i=1}^m \theta^{(i)} \xrightarrow{\text{a.s.}} \mathbb{E}(\theta | \mathbf{x}) \text{ as } m \rightarrow \infty$$

- Central limit theorem for the **accuracy**:

$$\bar{\theta}_{1:m} \sim N\left(\mathbb{E}(\theta | \mathbf{x}), \frac{\mathbb{V}(\theta | \mathbf{x})}{m}\right)$$

Simulation from marginals by selection



Multinomial model with Dirichlet prior

- **Categorical counts:** $\mathbf{y} = (y_1, \dots, y_C)$, where $\sum_{c=1}^C y_c = n$.
- y_c = number of observations in c th category. Brand choices.
- **Multinomial model:**

$$p(\mathbf{y}|\boldsymbol{\theta}) \propto \prod_{c=1}^C \theta_c^{y_c}, \text{ where } \sum_{c=1}^C \theta_c = 1.$$

- **Dirichlet prior:** $\boldsymbol{\theta} \sim \text{Dirichlet}(\alpha_1, \dots, \alpha_C)$

$$p(\boldsymbol{\theta}) \propto \prod_{c=1}^C \theta_c^{\alpha_c - 1}.$$

- **Marginal distributions**

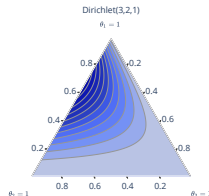
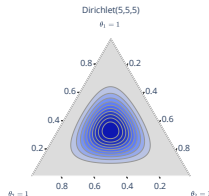
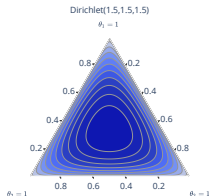
$$\theta_c \sim \text{Beta}(\alpha_c, \alpha_+ - \alpha_c), \text{ where } \alpha_+ = \sum_{c=1}^C \alpha_c$$

Dirichlet prior

$$(\theta_1, \dots, \theta_C) \sim \text{Dirichlet}(\alpha_1, \dots, \alpha_C)$$

$$\mathbb{E}(\theta_c) = \frac{\alpha_c}{\sum_{j=1}^C \alpha_j}$$

$$\mathbb{V}(\theta_c) = \frac{\mathbb{E}(\theta_c)(1 - \mathbb{E}(\theta_c))}{1 + \sum_{j=1}^C \alpha_j}$$



- 'Non-informative': $\alpha_1 = \dots = \alpha_K = 1$ (uniform and proper).

Multinomial model with Dirichlet prior

- **Simulation** from a $\text{Dirichlet}(\alpha)$ with $\alpha = (\alpha_1, \dots, \alpha_C)$:

```
Function RDIRICHLET( $\alpha$ )  
  for  $c$  in 1:C do  
    |  $y[c] \leftarrow \text{RGAMMA}(\alpha[c], 1)$   
  end  
  return  $y / \text{SUM}(y)$ 
```

- **Prior-to-Posterior:**

Multinomial data with Dirichlet prior

Model: $y | \theta \sim \text{Multinom}(n, \theta)$

Prior: $\theta \sim \text{Dirichlet}(\alpha)$, for $\alpha = (\alpha_1, \dots, \alpha_C)$

Posterior: $\theta | y \sim \text{Dirichlet}(\alpha + y)$

$y = (y_1, \dots, y_C)$ are counts in C categories

$\theta = (\theta_1, \dots, \theta_C)$ are category probabilities.

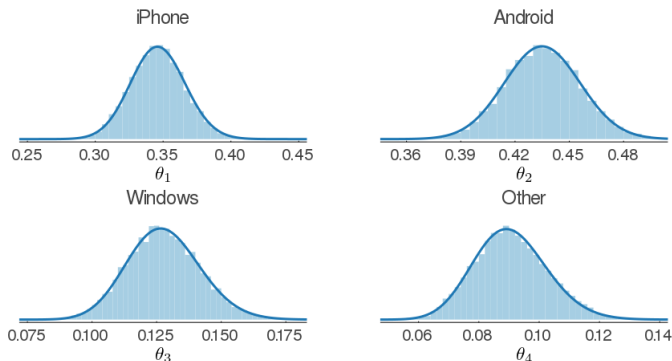
Example: smartphone market shares

- Survey among 513 smartphones owners:
 - ▶ 180 used mainly an iPhone
 - ▶ 230 used mainly an Android phone
 - ▶ 62 used mainly a Windows phone
 - ▶ 41 used mainly some other mobile phone.
- Old survey: iPhone 30%, Android 30%, Windows 20%, Other 20%.
- **Pr(Android has largest share | Data)**
- Prior: $\alpha_1 = 15, \alpha_2 = 15, \alpha_3 = 10$ and $\alpha_4 = 10$ (prior info is equivalent to a survey with only 50 respondents)
- Posterior: $(\theta_1, \theta_2, \theta_3, \theta_4) | \mathbf{y} \sim \text{Dirichlet}(195, 245, 72, 51)$.
- **R Notebook:** Multinomial.Rmd

Posterior simulation output

draw	θ_1	θ_2	θ_3	θ_4	I
1	0.33	0.47	0.10	0.09	1
2	0.34	0.44	0.11	0.09	1
3	0.36	0.41	0.13	0.08	1
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
10,000	0.35	0.43	0.14	0.08	1
Mean	0.34	0.43	0.13	0.09	0.99

Example: smartphone market shares



■ $\Pr(\text{Android has largest share} \mid \text{Data}) = 0.991$